

Data Structures and Algorithms – IT1170
Practical 1 – Introduction to Algorithms / Answers

Exercise 1

1. Navigation system can be considered as a real-life algorithm. List two similar examples.

- ATM Cash Withdrawal Process
- Online Food Ordering Process

2. Write a sample algorithm for a navigation system.

- Step 1: Enter destination.
- Step 2: Obtain current location.
- Step 3: Calculate shortest route.
- Step 4: Display route on map.
- Step 5: Give turn-by-turn directions.
- Step 6: Reach destination.
- Step 7: Stop navigation.

3. Write an algorithm to find the maximum of three numbers.

- Step 1: Start
- Step 2: Input A, B, C
- Step 3: Set Max = A
- Step 4: If B > Max, set Max = B
- Step 5: If C > Max, set Max = C
- Step 6: Display Max
- Step 7: End

4. Write an algorithm to find the average of N numbers.

- Step 1: Start
- Step 2: Input N
- Step 3: Set Sum = 0
- Step 4: Repeat N times
 - Input Number
 - Sum = Sum + Number
- Step 5: Average = Sum / N
- Step 6: Display Average
- Step 7: End

Exercise 2

5. Pseudocode to find maximum of three numbers

BEGIN

INPUT A, B, C

Max $\leftarrow$ A

IF B > Max THEN

Max $\leftarrow$ B

END IF

IF C > Max THEN

Max $\leftarrow$ C

END IF

DISPLAY Max

END

6. Pseudocode to find average of N numbers

BEGIN

INPUT N

Sum $\leftarrow$ 0

FOR i $\leftarrow$ 1 TO N

INPUT Number

Sum $\leftarrow$ Sum + Number

END FOR

Average $\leftarrow$ Sum / N

DISPLAY Average

END

Exercise 3

7. Java Program to Find Maximum of Three Numbers

```
import java.util.Scanner;

public class MaximumNumber {
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = input.nextInt();

        System.out.print("Enter second number: ");
        int b = input.nextInt();

        System.out.print("Enter third number: ");
        int c = input.nextInt();

        int max = a;

        if (b > max) {
            max = b;
        }

        if (c > max) {
            max = c;
        }

        System.out.println("Maximum number is: " + max);

        input.close();
    }
}
```